# Alok Prasad

+91 9262794486 | alokprasad573@gmail.com | LinkedIn | GitHub | Bhilai, Chhattisgarh

## CAREER OBJECTIVE

Computer Science undergraduate with skills in Python, MERN Stack, Machine Learning, and Generative AI. Built multiple AI and full-stack projects and seeking an entry-level software development or AI/ML opportunity.

## EDUCATION

**Rungta College of Engineering and Technology** Bhilai, Chhattisgarh
*Bachelor of Technology in Computer Science and Engineering (Artificial Intelligence)* Aug 2023 – May 2027
**CGPA: 7.0**

## TECHNICAL SKILLS

**Programming Languages**: Python, SQL
**AI/ML**: Machine Learning, Deep Learning, Computer Vision, NLP, Data Analysis, Generative AI, RAG
**Frameworks & Libraries**: TensorFlow, PyTorch, LangChain, Scikit-learn, NumPy, Pandas, FastAPI
**Databases**: MySQL, MongoDB, Supabase, Vector Databases
**Cloud & DevOps**: Git, GitHub, Docker, Kubernetes, CI/CD, AWS(Bedrock, S3, EC2, ElasticBeans)
**Web Technologies**: HTML, CSS, JavaScript, React.js, Tailwind CSS, Node.js, Express.js, REST APIs

## PROJECTS

**Solar Panel Anomaly Detection using Deep Learning** | Python, TensorFlow, OpenCV

- Developed an automated defect detection system for solar panels to reduce manual inspection efforts.
- Fine-tuned **EfficientNetV2B0** using transfer learning on a dataset of **870 images** across **6 defect classes**.
- Applied image preprocessing, resizing (**224 * 224**), and custom classification layers for multi-class prediction.
- Achieved **80% accuracy** and **0.92 F1-score** for snow defect detection, enabling predictive maintenance.

**GenAI-Based Health Insurance Auditor** | Python, LangChain, Groq LLM, Pinecone

- Built an AI-driven system to automate health insurance claim auditing and compliance verification.
- Developed a **RAG pipeline** for policy retrieval and audit report generation.
- Implemented semantic search using **Pinecone**, Sentence-Transformers, and **Llama 3.3** via Groq.
- Improved audit transparency and reduced manual claim processing effort through automated validation workflows.

**Intelligent AI Attendance with Face Recognition** | *Python, Streamlit, Scikit-Learn, Supabase*

- Developed a biometric attendance system to eliminate proxy attendance and manual record keeping.
- Built a facial recognition pipeline using **dlib**, **ResNet embeddings**, and a **KNN classifier**.
- Integrated **Supabase** for real-time attendance synchronization and developed portals using Streamlit.
- Reduced attendance errors by **90%** and administrative effort by **50%** through automation.

## CERTIFICATIONS

**MERN Full Stack Development** — ApnaCollege
**Data Science & AI Masters** — Udemy

## ACHIVEMENTS

- Secured **2nd Place** in a debugging competition.